Graph Sketching: Advanced Rational Functions

Questions 1–19

Question 1. Find the equations of the asymptotes of each of the following curves:

a) $y = \dfrac{x^2 - x + 2}{x + 1}$

b) $y = \dfrac{x^2 + 2x - 6}{x - 4}$

c) $y = \dfrac{3x^2 + x - 7}{x + 2}$

d) $y = \dfrac{11 - x^2}{x + 3}$

Question 2. Sketch the graphs of each of the following curves. You must clearly show any asymptotes:

a) $y = \dfrac{x^2 - 2x - 1}{x + 1}$

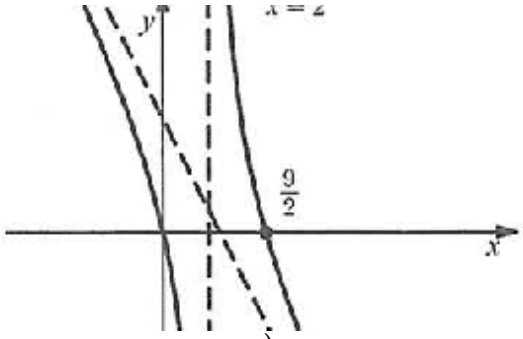
b) $y = \dfrac{2x^2 + 7x + 1}{x + 3}$

c) $y = \dfrac{6x^2 + x + 6}{1 - 3x}$

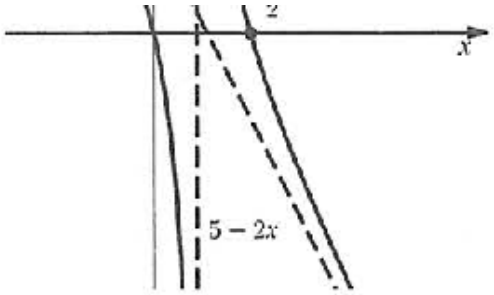
d) $y = \dfrac{51 + 5x - x^2}{x + 5}$

Question 3. Each of the following curves are of the form $y = \dfrac{ax^2 + bx + c}{x + d}$, where a, b, c, d are integers. Determine the values of a, b, c, d.

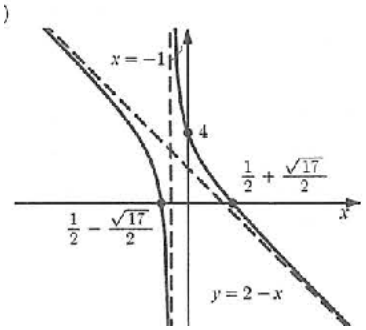
a)



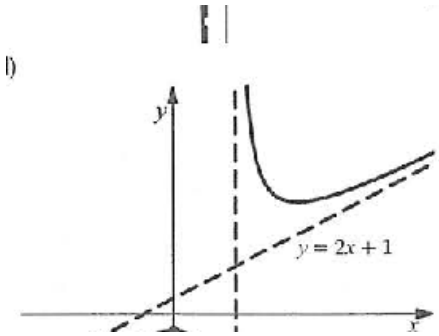
b)



c)



d)



Question 4. Find the equations of the asymptotes of each of the following curves, where:

a) $y = \dfrac{ax^2}{x + 1}$ for non-zero constant a

b) $y = \dfrac{ax^2}{x - 2}$ for non-zero constant a

c) $y = \dfrac{ax^2 + bx}{x - 1}$ for non-zero constants a and b

d) $y = \dfrac{ax^2 + b}{x + a}$ for non-zero constants a and b

e) $y = \dfrac{ax^2}{bx - c}$ for non-zero constants a, b, c

Question 5. One of the two asymptotes of the curve $y = \dfrac{x^2 + kx + 4}{x - 1}$, where k is a constant, is $y = x + 3$.

a) State the equation of the other asymptote.

b) Find the value of k.

Question 6. One of the two asymptotes of the curve $y = \dfrac{2x^2 + kx - 3}{x - 2}$, where k is a constant, is $y = 2x + 7$.

a) State the equation of the other asymptote.

b) Find the value of k.

Question 7. A curve has equation $y = \dfrac{ax^2 + bx + 4}{x - 1}$ and has asymptote $y = 3x + 2$.

a) Find the values of a and b.
b) Write down the equation of the other asymptote.
c) Sketch the curve.

Question 8. The function f is defined where $x \neq 1$ by $f(x) = \dfrac{x^2 + 4x + 7}{x + 1}$. The line $y = k$ intersects the curve.

a) Show that $(k + 2)(k - 6) \geq 0$.
b) Hence, find the coordinates of the stationary points of the curve $y = f(x)$.
c) Find the equations of the asymptotes of the graph of $y = f(x)$.
d) Sketch the graph of $y = f(x)$.

Question 9. The function f is defined where $x \neq -2$ by $f(x) = \dfrac{-x^2 + 6x + 12}{x + 2}$. The line $y = k$ intersects the curve.

a) Show that $(k - 6)(k - 14) \geq 0$.
b) Hence, find the coordinates of the stationary points of the curve $y = f(x)$.
c) Find the equations of the asymptotes of the graph of $y = f(x)$.
d) Sketch the graph of $y = f(x)$.

Question 10. A new residential district is being developed. The rate of change of the population P (in thousands) is modelled by

$$\frac{dP}{dt} = \frac{2t^2 - t - 3}{t + 2}, \qquad t \geq 0,$$

where t is the time in years after construction begins.

a) Sketch the curve of $\dfrac{dP}{dt}$ for $t \geq 0$.
b) With reference to the rate of change of the population, explain what is happening to the population according to the model.
c) Find an expression for $P(t)$.
d) Given that initially there are 4 thousand residents, find the particular solution.
e) Determine the population after 4 years.

Question 11. A curve has equation $y = \dfrac{ax^2 + bx + 7}{x + 3}$ and has asymptote $y = x + 1$.

a) Find the values of a and b.
b) Write down the equation of the other asymptote.
c) Without using Calculus, find the coordinates of the turning points.
d) Sketch the curve.

Question 12. A curve has equation $y = \dfrac{x^2}{x + 2}$.

a) Find all the equations of the asymptotes of the curve.
b) Without using Calculus, find the coordinates of the stationary points of the curve.
c) Sketch the curve.

The region bounded by the curve, the x-axis and the lines $x = 0$ and $x = 2$ is rotated about the x-axis.

a) Show that the volume formed is given by $V = \pi\left(\dfrac{68}{3} - 32\ln 2\right)$ units^3.

Question 13. A company manufactures wireless ear buds. The average production cost per batch C (in hundreds of pounds), when producing x batches per day, is modelled by

$$C = \frac{3x^2 + 4x + 2}{x + 1}.$$

a) Find all the equations of the asymptotes of the curve.
b) Without using Calculus, find the coordinates of the stationary points of the curve, giving your answers to three decimal places.
c) Sketch the curve.
d) Find the mean value of the cost function over the production interval $1 \leq x \leq 4$.
e) Interpret your answer in the context of the manufacturing process.

Question 14. Find the equations of the asymptotes of each of the following curves:

a) $y = \dfrac{x^3 - 4}{x + 2}$
b) $y = \dfrac{x^3 + 3x^2 + 1}{x - 2}$
c) $y = \dfrac{1 - x^3}{x + 1}$

Question 15. Sketch the graphs of each of the following curves. You must clearly show any asymptotes:

a) $y = \dfrac{x^3 + x^2 - 4}{x + 1}$

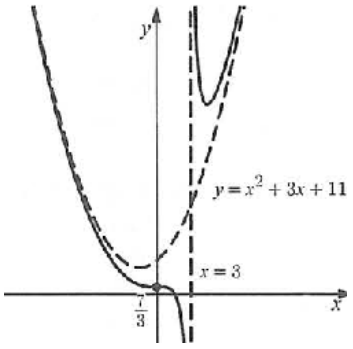
b) $y = \dfrac{18 + x - 3x^2 - x^3}{x + 3}$

c) $y = \dfrac{x^3 - 3x^2 - 4x + 12}{x + 1}$

d) $y = \dfrac{3x^3 + 4x^2 + x + 1}{x - 2}$

Question 16. Each of the following curves are of the form $y = \dfrac{ax^3 + bx^2 + cx + d}{x + p}$, where a, b, c, d, p and q are integers. Determine the values of a, b, c, d, p.

a)



Question 17. Find the equations of the asymptotes of each of the following curves, where:

a) $y = \dfrac{ax^3}{x+1}$ for non-zero constant a

b) $y = \dfrac{ax^3}{x-2}$ for non-zero constant a

c) $y = \dfrac{ax^3+bx^2}{x+1}$ for non-zero constants a and b

d) $y = \dfrac{ax^3+bx}{x^2+a}$ for non-zero constants a and b

e) $y = \dfrac{ax^3}{bx-c}$ for non-zero constants a, b, c

Question 18. One of the two asymptotes of the curve $y = \dfrac{x^3+kx^2-2}{x-1}$, where k is a constant, is $y = x^2+5x+5$.

a) State the equation of the other asymptote.

b) Find the value of k.

Question 19. One of the two asymptotes of the curve $y = \dfrac{x^3 - kx^2 + 47}{x + 3}$, where k is a constant, is $y = x^2 - 5x + 15$.

a) State the equation of the other asymptote.

b) Find the value of k.